RTU Imp topic for 4th Semester

DMS:-

unit 1 Question based on inclusion and exclusion property in 4 marks and (10 marks)

Equivalence Relation
types of function
mathematical induction rule 4 marks and (10 marks)

unit 2 Propositional logic - totology , contradiction , quantifiers,

question based on prove two propositional logic
CNF and DNF
Finite state machine

unit 3 POSET , hasse diagram - lower and upper bound

Lattice and it's property
bounded , complemented lattice and binomial theorem formula
recurrence relation solution for homogenous or non homogenous (10 marks)
generating function in (4 or 10 marks)

unit 4 Binary operation Alegabric properties - commutative, closure, associative, identity and inverse

group ,semi group and monoid for prove
Abelian group : prove 10 marks
Normal group : prove
ring field 10 marks
Isomorphism and homomorphism of group

unit 5 definition of different graphs

Hamiltion and eulerian graph
Isomorphism and homomorphism of graph
chromatic number (2 marks but imp)
Dijkshtra algorithm for shortest path (10 marks)